3-13-26 (WEEK 3)

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Change notation to have "$\mathfrak{a}$" in place of "I", but continued

LEMMA

For any nonzero $\mathfrak{a} \trianglelefteq A$, there are max ideals P_i s.t. $\boxed{\mathfrak{a} \supseteq \prod_{i=1}^{r} P_i}$ $(\triangle)$

Proof

If $\mathfrak{a}$: max, then it's done.

Assume $\mathfrak{a}$: not max, then $\mathfrak{a}$: not prime ideal ($\because$ DDK)

$\therefore$ By def, $\exists\, x, y \in A \setminus \mathfrak{a}$, s.t. $xy \in \mathfrak{a}$

$\Rightarrow (\mathfrak{a}+(x)), (\mathfrak{a}+(y)) \supsetneq \mathfrak{a}$ and $(\mathfrak{a}+(x))(\mathfrak{a}+(y)) \subseteq \mathfrak{a}$

If both $(\mathfrak{a}+(x))$ and $(\mathfrak{a}+(y))$ satisfy $(\triangle)$, then so must $\mathfrak{a}$.

If $\mathfrak{a}$ does NOT satisfy $(\triangle)$, then one of $(\mathfrak{a}+(x)), (\mathfrak{a}+(y))$ does not satisfy $(\triangle)$. WLOG, say it's $(\mathfrak{a}+(x))$. Then, $(\mathfrak{a}+(x)) \supsetneq \mathfrak{a}$

Repeat this process + A is noeth $\Rightarrow \mathfrak{a} \supseteq \prod_{i=1}^{r} P_i$ $\square$

DEFINITION

(i) For $(0) \neq I \trianglelefteq A$, we define $I^{-1} := \{x \in \mathrm{Frac}(A) \mid xI \subseteq A\}$

(ii) A fractional ideal of A is a nonzero f.g. A-mod $I \subseteq \mathrm{Frac}(A)$ s.t. $(0) \neq I := \{x \in \mathrm{Frac}(A) \mid xI \subseteq A\}$

f.g. : $I \neq (0)$

LEMMA

For a nonzero ideal $\mathfrak{a} \trianglelefteq A$, if $\alpha \in \mathrm{Frac}(A)$ s.t. $\alpha\mathfrak{a} = \mathfrak{a}$, then $\alpha \in A$

Proof

A is noeth $\Rightarrow \mathfrak{a}$: f.g. A-mod

Let $\{e_1, \dots, e_n\}$ be a set of generators of $\mathfrak{a}$

Then, $\forall i$, $\alpha e_i = \sum_{j=1}^{n} c_{ij} e_j$

$$\Rightarrow \begin{pmatrix} \alpha & & \\ & \ddots & \\ & & \alpha \end{pmatrix} \begin{pmatrix} e_1 \\ \vdots \\ e_n \end{pmatrix} = \underbrace{(c_{ij})}_{C} \begin{pmatrix} e_1 \\ \vdots \\ e_n \end{pmatrix}$$

$$\Rightarrow (C - \alpha I) \begin{pmatrix} e_1 \\ \vdots \\ e_n \end{pmatrix} = 0$$

$\therefore \det(C - \alpha I)\, e_i = 0 \quad \forall\, 1 \leq i \leq n$

As $\mathfrak{a}$: nonzero, $\exists\, e_i \neq 0$ for some i.

As A is a domain, thus $\det(C - \alpha I) = 0$, i.e. α is a root of $\det(C - xI) = 0$

$\therefore \alpha$ is integral

As A is int closed (DDK), thus $\alpha \in A$ $\square$

LEMMA

Let $P \trianglelefteq A$ be maximal, then

① $P^{-1} \supsetneq A$

② $PP^{-1} = A$

Proof

① Clearly, $A \subseteq P^{-1}$

Let $0 \neq \alpha \in P$, then $P \supseteq (\alpha) \supseteq \prod_{i=1}^{r} P_i$ with P_i: max and r: min.

(Note: $P = P_i$ for some i, $1 \leq i \leq r$. Otherwise, P and P_i are max, $x_i \in P_i \setminus P$, then $x_1 \cdots x_r \in P$. WLOG, assume $P = P_1$)

$\forall \beta \in \prod_{i=2}^{r} P_i$, $\beta P_1 \subseteq (\alpha) \subseteq A \Rightarrow \frac{\beta}{\alpha} P_1 \subseteq A \Rightarrow \frac{\beta}{\alpha} \in P_1^{-1}$ $\forall \beta \in \prod_{i=2}^{r} P_i$

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If $P^{-1}=A$, then $\frac{\beta}{\alpha}\in A \Rightarrow \beta\in(\alpha) \Rightarrow \prod_{i=2}^{r} P_i \subseteq (\alpha)$ $\times$
$\therefore P^{-1} \supsetneq A$ □

② By def, $PP^{-1}\subseteq A$ and is an ideal, satisfying $P\subseteq PP^{-1}\subseteq A$
As P: max, $PP^{-1}=P$ or $PP^{-1}=A$
By Lemmas, $PP^{-1}\neq P$
$\therefore PP^{-1}=A$ □

PROPOSITION

Let $a\trianglelefteq A$. Assume $\exists$ maximal ideals P_i, Q_j s.t. $a=\prod_{i=1}^{n}P_i=\prod_{j=1}^{m}Q_j$. Then, $m=n$ and Q_j is a rearrangement of P_i

Proof

Consider induction on $\min\{n,m\}$.

- $\min\{n,m\}=1$: WLOG, let $n=1$, then $P_1=\prod_{j=1}^{m}Q_j$. Thus, $Q_j=P_1$ for some $1\leq j\leq m$. WLOG, $j=1$. If $m\geq 2$, $A=\prod_{i=2}^{m}Q_i$ $\times$ $\therefore m=1$ ✓
- $\min\{n,m\}>2$: Similarly, we can assume $P_1=Q_1 \Rightarrow \prod_{i=2}^{n}P_i=\prod_{j=1}^{m}Q_j$. As $\min\{n-1,m-1\}<\min\{n,m\}$, we can conclude by ind hyp □

LEMMA

Let $a\trianglelefteq A$ be nonzero and $P\trianglelefteq A$ be maximal
If $P\supseteq a$, then $a=Pa'$ with $a'\trianglelefteq A$ and $a'\supsetneq a$

Proof

Think "$a=P(P^{-1}a)$" so we first need $P^{-1}a\subseteq a$

Notice, $A=PP^{-1}\supseteq P^{-1}a$, where $P^{-1}a$: ideal
If $P^{-1}a=a$, then $P^{-1}\subseteq A$ $\times$
$\therefore P^{-1}a\supsetneq a$ □

RECALL THEOREM

Given A: Dedekind domain, then $(0)\neq I\trianglelefteq A$, there are proper max ideals $P_i\trianglelefteq A$ s.t. $a=\prod_{i=1}^{n}P_i$ and this decomposition is unique up to rearrangement.

Proof

If a: max, we are done.
If not, $\exists P_1\trianglelefteq A$, P_1: max, s.t. $P_1\supsetneq a \Rightarrow a=P_1a_1$, $a\subsetneq a_1\subsetneq A$

If a_1 is a product of max ideals, we are done.
If not, repeat the decomposition.

The process terminates only when a_i: product of max ideals
If this never stops, then we have an ascending chain $a\subsetneq a_1\subsetneq \dots$ since A: noeth □

COROLLARY

Say $a=\prod_{i=1}^{r}P_i^{e_i}$, P_i: distinct max, $a\supseteq I$ with nonzero $I\trianglelefteq A$
Then, $I=\prod_{i=1}^{r}P_i^{f_i}$, $f_i\geq e_i$

THEOREM

Let A: Dedekind domain, $K:=\mathrm{Frac}(A)$
If K'/K: finite sep ext, and $A':=$ int closure of A in K', then A': Dedekind domain

Proof ($\mathbb{Z}\rightsquigarrow \mathcal{O}_{K'}$, $K=\mathbb{Q}$)

① "A' is noetherian"
As K'/K: sep + finite, say $K'=K(\theta)$ for some $\theta\in K'$ by Primitive Element Thm
We can choose $\theta\in A'$

Using the Tr form technique, $\exists d\in A$, s.t. $A'\subseteq \bigoplus_{i=0}^{n-1}A\frac{\theta^i}{d} \cong A^n$, $n=[K':K]$
As A: noeth, thus A^n: noeth $\Rightarrow A'\subseteq A^n$: noeth

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② "Nonzero prime ⇒ max ideal"

Say $\mathfrak{Q} \trianglelefteq A'$ is a nonzero prime

We know $\mathfrak{Q} \cap A$: prime, and $\mathfrak{Q} \cap A$: nonzero (Otherwise, $\overset{A'}{\overset{\cup}{(0)}} \cap A = \mathfrak{Q} \cap A = (0)$ contradicts A'/A: integral)

A: DDK ⇒ $\mathfrak{Q} \cap A$: max

A'/A: integral, so $\mathfrak{Q}$: max ⇔ $\mathfrak{Q} \cap A$: max

③ "Integrally closed": Like the "$\mathcal{O}_K$ case" □

DECOMPOSITION OF $p\mathcal{O}_K$

IMPORTANT TOOLS (For $\mathcal{O}_K$: ring of integers)

DEFINITION

For $(0) \neq \mathfrak{a} \trianglelefteq \mathcal{O}_K$, $\underbrace{\mathrm{Nr}}_{\text{norm}}(\mathfrak{a}) := \#(\mathcal{O}_K/\mathfrak{a}) \underbrace{<}_{\mathcal{O}_K \text{ special property}} \infty$

PROPOSITION

Say $\mathfrak{a} = (\alpha) \trianglelefteq \mathcal{O}_K$ with $\alpha \neq 0$, then $\mathrm{Nr}(\mathfrak{a}) = |\mathrm{Nr}_{K/\mathbb{Q}}(\alpha)|$

Proof

By Structure Thm of mod over PID,

$(\alpha) = \bigoplus_{i=1}^{n} \mathbb{Z} b_i e_i \subseteq \mathcal{O}_K = \bigoplus_{i=1}^{n} \mathbb{Z} e_i$, $b_i \in \mathbb{Z}_{>0}$

Define $m_\alpha : K \longrightarrow K$, $x \longmapsto \alpha x$ $\rightsquigarrow$ $m_\alpha : \mathcal{O}_K \longrightarrow \mathcal{O}_K$

$\mathrm{coker}(m_\alpha) = \mathcal{O}_K/(\alpha)$

Then, for $\alpha \in \mathcal{O}_K$, we have $m_\alpha(\mathcal{O}_K) \subseteq \mathcal{O}_K$

Let M_α be the matrix rep w.r.t. $\{e_i\}$.

Then, $\alpha e_i = \sum_{r=1}^{n} c_{ri} e_r$, where $\mathrm{Nr}_{K/\mathbb{Q}}(\alpha) = \det(M_\alpha)$

We have $M_\alpha \begin{pmatrix} 0 \\ \vdots \\ 1 \\ \vdots \\ 0 \end{pmatrix} \leftarrow i\text{th} = \begin{pmatrix} c_{1i} \\ \vdots \\ c_{ni} \end{pmatrix}$

$\because$ $\{\alpha e_i\}$ and $\{b_i e_i\}$ are both bases of $(\alpha) = \mathfrak{a}$

Thus, we also have $b_i e_i = \sum_j (\alpha e_j) d_{ji}$, where d_{ji} must be invertible since αe_i can be written similarly.

$\Downarrow$ $\det = \pm 1$ since it's in $\mathbb{Z}$

$\Rightarrow M_\alpha(d_{ji}) = \begin{pmatrix} b_1 & & \\ & \ddots & \\ & & b_n \end{pmatrix}$

$\Rightarrow \underbrace{\prod_{i=1}^{n} b_i}_{\#(\mathcal{O}_K/\mathfrak{a}) = \mathrm{Nr}_{K/\mathbb{Q}}(\alpha)} = \underbrace{\det(M_\alpha)}_{} \underbrace{\det(d_{ji})}_{\{\pm 1\}}$ □